

Frederick Khasanto

Research Profile | PhD in Intelligent Transportation, HKUST (Guangzhou)

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CURRENT DOCTORAL DIRECTION

PhD in Intelligent Transportation, The Hong Kong University of Science and Technology (Guangzhou), 2026 - present

Advisor: Prof. Benedict Jun MA

I am beginning doctoral research in Intelligent Transportation while building on a background in data science, machine learning, language, and multimodal AI.

RESEARCH EXPERIENCE

Offensive Speech Detection - Research Assistant, CUHK(SZ)

Transformer-based hate-speech literature review and dataset collection for supervised classification.

Vision-Language Models - Summer Research Intern, VLAA Lab, UC Santa Cruz

Reviewed multimodal architectures including LLaVA and LLaVA-CoT under the supervision of Hardy Chen and Prof. Cihang Xie.

SELECTED PROJECTS

Project	Role / Method	Outcome
Diffusion Models for X-ray Image Generation	Model implementation / DDPM, PyTorch	Medical X-ray generation
KnowYourBody	Web app & LLM integration / LLM APIs	Educational health analytics
SmartBot	Chatbot & cloud workflow / AWS Cloud Functions	Clinic appointment scheduling

TEACHING

Teaching Fellow experience in Programming Methodology, Applied Machine Learning, and Probability & Statistics II.

RESEARCH INTERESTS & SKILLS

Interests: Intelligent Transportation, Machine Learning, Data Science, NLP, Multimodal AI

Tools: Python, R, C++, SQL, PyTorch, Scikit-learn, Pandas, NumPy, Git, LaTeX